Hypercube geodesics with few colour changes

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Abstract

What is the maximum, over all 2-colourings of the edges of the n-dimensional hypercube Q_n, of the minimal number of times a path between a vertex v and its antipode $\bar{v}$ changes colour? A conjecture of Norine, in a form due to Feder and Subi, states that this maximum should be 1. The previous best-known upper bound on the number of colour changes was $(\frac{5}{16} + o(1))n$ due to Kirchweger, Peitl, Subercaseaux, and Szeider. We improve this bound and answer a question of Leader and Long by finding a geodesic path with at most $(\frac{\pi}{2} + o(1))\sqrt{n}$ colour changes. In fact, we show that this is the expected number of colour changes for a uniformly random start vertex. This is optimal (up to the constant) when the start vertex is chosen uniformly at random.

This is the "annotated" form of this paper. There will be notes like this one in this margin to provide informal commentary of what's going on.

Note that the results of this paper have been Lean-formalised with the help of AI – see my website.

1 Introduction

The hypercube is a central object in combinatorics, and many problems concerning the structures that can be found on taking subsets or colourings of the edges have been studied. While a result of Alon, Radoičić, Sudakov, and Vondrák [1] precisely characterises those graphs for which a monochromatic copy will appear in a sufficiently large k-coloured hypercube Q_n (for k fixed), much remains unknown about structures which grow with the dimension n, or if one only wants approximately monochromatic structures. One example of such a problem is a conjecture of Norine (see [10]), who asked whether every antipodal 2-colouring of the hypercube Q_n (a colouring where antipodal edges receive different colours) contains a monochromatic path between some pair of antipodal vertices. This conjecture was modified by Feder and Subi [5] to the following.

Conjecture 1.1. *Every 2-colouring of $E(Q_n)$ contains a path between some pair of antipodal vertices which changes colour at most once.*

Note that an AI-assisted topological proof of Conjecture 1.1 has been found since I wrote this paper.

It was proved in [5] that one can find a monochromatic path between some pair of vertices at Hamming distance at least $\lceil n/2 \rceil$, and thus there is a pair of antipodal vertices and a path between them with at most $\lfloor n/2 \rfloor$ colour changes. Long [9] proved that subgraphs of Q_n of large average degree must contain exponentially long paths, and later Leader and Long [8] showed that one can also find geodesics of length $\lceil n/2 \rceil$ in such subgraphs, proving the existence of a geodesic between antipodes with at most $\lfloor n/2 \rfloor$ colour changes. They moreover conjectured that the path in Conjecture 1.1 can be taken to be a geodesic and posed the following, weaker, question.

Basically all of these questions can be sensibly asked for paths or for geodesics.

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Question 1.2. *Is it true that for every 2-colouring of $E(Q_n)$, there exist two antipodal vertices x and x' that are joined by a path that changes colour $o(n)$ times?*

Our main result is the following, which provides an affirmative answer to Question 1.2.

Theorem 1.3. *For any 2-colouring of the edges of the hypercube Q_n, there is a vertex $v \in V(Q_n)$ and a geodesic path from v to its antipode $\bar{v}$ which changes colour at most $(\frac{\pi}{2} + o(1))\sqrt{n}$ times.*

The antipode of a vertex v is denoted $\bar{v}$ throughout.

Indeed, this continues the pattern of geodesic paths being sufficient. We note that Theorem 1.3 constitutes the first sub-linear bound on this problem. Before our result, Dvořák [4] used a careful consideration colourings of copies of $Q_3 \subseteq Q_n$, proved that there always exists a pair of antipodal vertices and a geodesic between them with at most $(\frac{3}{8}+o(1))n$ colour changes. In a different direction, Conjecture 1.1 is also known to hold for values of n up to 8 [11, 12, 6, 7] by application of computer search with SAT-solvers.

This also took a random start point, but because they broke Q_n up into Q_3-copies, this method seems stuck at proving linear bounds.

Combining these two approaches, Kirchweger, Peitl, Subercaseaux, and Szeider [7] found a more efficient encoding of the problem, allowing them to not only prove the $n = 8$ case of the conjecture, as mentioned above, but to also use an automated analysis of Q_6 to improve Dvořák's bound to $(\frac{5}{16} + o(1))n$ colour changes.

Question 1.2 was repeated by Ivan and Johnson as part of a larger collection of open problems [2]. They therein raised the question of what can be said if we insist that the vertex v at the start of the geodesic is chosen uniformly at random in the hypercube, as follows.

Somewhat related to this: it turns out that dealing with a random *colouring* is easy: the greedy algorithm gives a monochromatic geodesic with positive probability. It's not immediately obvious that this works w.h.p.

Question 1.4. *How large can be the minimum number of colour-changes on a path between a randomly chosen pair of antipodal vertices? In other words, determine the maximum (over all edge colourings of Q_n) of the quantity*

$$\frac{1}{2^n} \sum_{v \in Q_n} (\textit{the minimum number of colour changes on a } v\textit{-to-}\bar{v} \textit{ path}) \tag{1.1}$$

Our proof chooses the end of the geodesic uniformly at random and – as will be discussed in the concluding remarks – there is a colouring of Q_n where the value in (1.1) is $\Omega(\sqrt{n})$. This provides a strong resolution of Question 1.4 for geodesics as well as more general paths.

In fact, the constant in the lower bound is $1/\sqrt{\pi}$, so the upper and lower bound differ by a factor of less than 3, which is pretty tight as things go.

1.1 Intuition and proof outline

We prove that there is a choice of vertex v and a geodesic path from v to $\bar{v}$ with few colour changes. Our starting vertex v is chosen uniformly at random in $V(Q_n)$, but even so we must be careful in our choice of path: if we attempt to pick a geodesic from v to $\bar{v}$ greedily, there is no immediate way of ruling out the possibility of running into lots of "dead-end" vertices – vertices x where our geodesic path arrives via, say, a red edge, but all edges from x away from v are blue, and so we are forced to change colour. A key intuition for our proof is the following informal heuristic, which says that arriving at such a dead end should be very unlikely.

If we're not careful, we might "use up" this relatively small amount of randomness in just getting some nice properties around the vertex v, and then we have to fight against an adversarial colouring (which doesn't go well)

Fix a vertex x, and consider picking a vertex v uniformly at random amongst all vertices at a distance k from x, and assume that we want our geodesic to start from v and go through x. The only way in which x could be a dead end is if either all of its red neighbours (that is, neighbouring vertices in Q_n connected by a red edge) or all of its blue neighbours are closer to v than x is. However, if x is in many red edges and many blue edges and k is not too close to n, then this is very unlikely to happen. If on the other hand x has very few edges of one colour, say, blue, then while it does have a non-negligible chance of being a dead end in blue, most paths arriving at x use a red edge, from which there are plenty of options for continuing the geodesic path with another red edge.

We'll end up thinking about paths between two random endpoints (at a fixed distance), and we'll sometimes think of one end as fixed, and sometimes the other.

When we do the analysis, the inequalities will end up being tight for close-to-balanced vertices.

To formalise this intuition, we consider the function $f_c(x, y)$, which is the minimum number of colour changes in a geodesic between vertices x and y ending in colour c. We in fact compute a weighted average of this function around the vertex y, weighted by the proportion of the edges around y in each colour. In fact, we compute several version of such an average, and relate them to each other in sequence (see Figure 1 for a pictorial representation of these averages).

Note that, if x and y are neighbours and the edge between them is blue, then $f_{\mathsf{Blue}}(x, y) = 0$ (of course), but also $f_{\mathsf{Red}}(x, y) = 1$: we can take the blue edge and then "change colour at y".

The crux of the argument is to step from weighting by the edges from y towards x to weighting by the edges away from x. But, if y is fixed and x is chosen uniformly at random at a specific distance from y, then the number of red edges from y towards x is distributed as a hypergeometric random variable, which is very easy to control. Linearity of expectation then allows us to sum these changes over a random path from x to $\bar{x}$, and the result follows.

2 Proof of the main theorem

We first give a complete description of all the notation that we will use throughout the proof, and then use this notation to state our key lemma, Lemma 2.2, in Section 2.1 before proceeding to give a proof of this lemma in Section 2.2.

2.1 Notation and reduction to a key lemma

Throughout this section we work on the graph Q_n of the hypercube in n dimensions with a colouring $\chi\colon E(Q_n) \to \{\text{Red}, \text{Blue}\}$. The value of n will be fixed throughout. Denote the vertex set and edge set of Q_n as V and E respectively. We write $x \sim y$ to mean that vertices x and y are adjacent, and $x \sim_c y$ to mean that they are connected by an edge with colour c. We call the two colours Red and Blue.

This is all standard notation, nothing new here.

We write $d(x, y)$ for the Hamming distance from x to y (i.e. the graph distance in Q_n), so for example $x \sim y$ implies $d(x, y) = 1$, and we write

$$N^{(k)}(x) \coloneqq \{y \in V : d(x, y) = k\}$$

for the kth neighbourhood of x.

We now introduce our non-standard notation. For all $x, y \in V$ and $c \in \{\text{Red}, \text{Blue}\}$, we define the sets I and J, which can be thought as the edges from a vertex respectively towards and away from the start of the path that we construct.

And now it gets non-standard. This is mostly book-keeping, except for the numbered equations.

Definition.

$$\begin{aligned} I_c^x(y) &:= \{z \in V : \ z \sim_c y,\ d(x,z) = d(x,y) - 1\}, \\ I^x(y) &:= \{z \in V : \ z \sim y,\ d(x,z) = d(x,y) - 1\}, \\ J_c^x(y) &:= \{z \in V : \ z \sim_c y,\ d(x,z) = d(x,y) + 1\}, \quad \text{and} \\ J^x(y) &:= \{z \in V : \ z \sim y,\ d(x,z) = d(x,y) + 1\}. \end{aligned}$$

I like to think of $I_c^x(y)$ as the "incoming edges" (from x to y) in colour c; we normally think of x as fixed and y as varying. $J_c^x(y)$ is then of course the "(j)outgoing edges".

We then define the following number, which will be explained in Section 2.2.

Think of $S^x(y)$ as measuring how much the distribution of colours changes between the incoming and outgoing edges at y (relative to x). It turns out that we'll be able to control this very precisely.

Definition.

$$S^x(y) := \left| \frac{|J_{\text{Red}}^x(y)|}{n - d(x,y)} - \frac{|I_{\text{Red}}^x(y)|}{d(x,y)} \right| = \left| \frac{|J_{\text{Blue}}^x(y)|}{n - d(x,y)} - \frac{|I_{\text{Blue}}^x(y)|}{d(x,y)} \right|. \tag{2.1}$$

For now we just note that the above equality comes from the fact that

$$|J_{\text{Red}}^x(y)| + |J_{\text{Blue}}^x(y)| = n - d(x,y) \quad \text{and} \quad |I_{\text{Red}}^x(y)| + |I_{\text{Blue}}^x(y)| = d(x,y).$$

We now discuss our use of random variables. We select two points $u, v \in V$ independently uniformly at random. $\mathbb{E}$ will be the unconditioned expectation, and for $k \in [n]$ (where $[n] = \{1, 2, \dots, n\}$), we will use the shorthand

The notation $\mathbb{E}_k$ thus implicitly depends on the two random variables u and v – these are the only names we will use for random vertices.

Definition.

$$\mathbb{E}_k[\,\cdot\,] := \mathbb{E}[\,\cdot \mid d(u,v) = k].$$

Definition. Write $f_c(x,y)$ for the minimum number of colour changes on a geodesic path from x to y ending in colour c. We allow the path to change colour at y, even if no edge of that colour appears in the path; for example, if $x \sim_{\text{Blue}} y$, then $f_{\text{Blue}}(x,y) = 0$ and $f_{\text{Red}}(x,y) = 1$.

We record a simple observation about these values.

e.g. just take the path from x to y ending in, say, blue, and then change to red at y. Thus $f_{\text{Red}}(x,y)$ is at most $f_{\text{Blue}}(x,y) + 1$.

Observation 2.1. *For all vertices $x, y \in V$, we have*

$$\left| f_{Red}(x,y) - f_{Blue}(x,y) \right| \le 1.$$

This follows from our ability to swap between colours at y if we wish.

Definition. We now define three weighted averages, which will be the central objects of our proof.

$$f(x,y) := \frac{1}{n-d(x,y)} \sum_{z\in J^x(y)} f_{\chi(yz)}(x,y) = \frac{1}{n-d(x,y)} \sum_{c\in\{\text{Red},\text{Blue}\}} |J^x_c(y)|\, f_c(x,y), \tag{2.2}$$

$$g(x,y) := \frac{1}{d(x,y)} \sum_{z\in I^x(y)} f_{\chi(yz)}(x,z), \quad \text{and} \tag{2.3}$$

$$h(x,y) := \frac{1}{d(x,y)} \sum_{z\in I^x(y)} f_{\chi(yz)}(x,y) = \frac{1}{d(x,y)} \sum_{c\in\{\text{Red},\text{Blue}\}} |I^x_c(y)| f_c(x,y). \tag{2.4}$$

Just look at the picture to make sense of these; it's a bunch of notation hiding a relatively simple idea. These are mostly (despite appearances) for notational convenience only.
The second parts of the equality follow from grouping edges by colour, as if x, c are fixed then $f_c(x,y)$ depends only on the colour of edge xy.

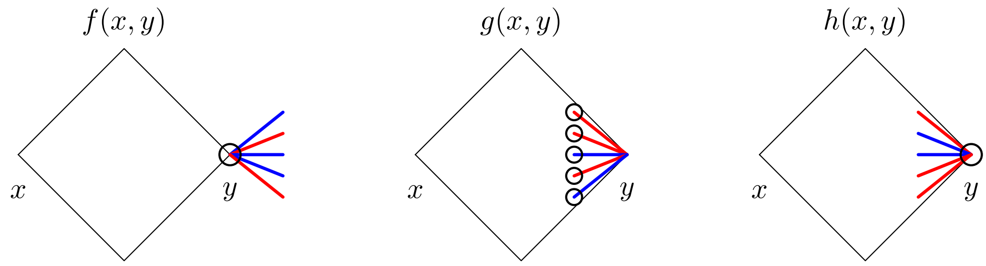

Figure 1: A pictorial representation of the functions f, g, and h. Each function averages $f_c(x,z)$ (the minimum number of colour changes on a geodesic path from x to z ending in colour c) over the edges shown in colour, where c is the colour of the edge and z is the circled vertex. The large diamond represents the sub-cube between x and y.

We are now ready to state our key lemma, which is as follows.

The big scary equation here is basically just "the thing which drops out of the calculations"; we'll end up just approximating it all down to depend only on n in the end.

Lemma 2.2. *Fix an integer $k \in [n]$. We have*

$$\mathbb{E}_k\big[f(u,v)\big] \le \frac{1}{2}(1+o(1)) \sum_{j=1}^{k} \sqrt{\frac{j(n-j)}{n}} \Big(\frac{1}{n-j} + \frac{1}{j}\Big). \tag{2.5}$$

Once we have Lemma 2.2, it is easy to deduce Theorem 1.3:

Proof of Theorem 1.3 assuming Lemma 2.2. We compute an upper bound on $\mathbb{E}_n[f(u,v)]$, which is the expected optimal number of colour changes between u and v which are conditioned to be antipodal.

We just approximate the sum by an integral, nothing difficult or lossy going on here.

$$\frac{1}{2}\sum_{j=1}^{n-1} \sqrt{\frac{j(n-j)}{n}} \Big(\frac{1}{n-j} + \frac{1}{j}\Big) = \sum_{j=1}^{n-1} \sqrt{\frac{n-j}{nj}} \le \int_0^{n-1} \sqrt{\frac{n-x}{nx}}\,\mathrm{d}x \le \sqrt{n}\int_0^1 \sqrt{\frac{1-y}{y}}\,\mathrm{d}y = \frac{\pi}{2}\sqrt{n}.$$

We have used in the above that the function $\sqrt{(n-x)/(nx)}$ is decreasing in x for $x > 0$. There is thus a choice of v for which there is a path from v to $\bar{v}$ at most the above number of colour changes, from which Theorem 1.3 follows immediately. $\square$

2.2 Proof of the key lemma

We prove Lemma 2.2 by induction on k. Indeed, the base case $k = 0$ is trivial as $f(v,v) = 0$ and the sum in (2.5) is over no terms, so is also zero, so it remains to prove the inductive step. The following three simple claims form the engine of our proof: we step from $\mathbb{E}_{k-1}[f(u,v)]$ to $\mathbb{E}_k[g(u,v)]$, then to $\mathbb{E}_k[h(u,v)]$, and finally to $\mathbb{E}_k[f(u,v)]$, completing the induction.

These three claims are where the real work is going on. Yet, somehow, there's actually almost no work at all. Perhaps the real work was actually writing down the definitions and setting up the random variables to allow these to go through.

Claim 2.3. *Fix an integer $k \in [n]$. We then have $\mathbb{E}_{k-1}[f(u,v)] = \mathbb{E}_k[g(u,v)]$.*

Proof. This claim in fact holds even if we condition on the value of u. Indeed, let $E_k(u)$ be the set of edges from $N^{(k-1)}(u)$ and $N^{(k)}(u)$. Both of the expectations in Claim 2.3 are equal to

$$\frac{1}{|E_k(u)|} \sum_{e \in E_k(u)} f_{\chi(e)}(u, v_e),$$

where v_e is the vertex of e in $N^{(k-1)}(u)$. Indeed, this follows immediately from the observation that we may count $E_k(u)$ by selecting a vertex $x \in N^{(k-1)}(u)$ and a neighbour y of x in $N^{(k)}(u)$, or by first selecting y in $N^{(k)}(u)$, and then selecting a neighbour x of y in $N^{(k-1)}(u)$. □

This proof is just noticing that both of these values count the same function on the same set of edges.

Claim 2.4. *For all $x, y \in V$, we have $h(x,y) \le g(x,y)$.*

Proof. Observing the definitions (2.3) and (2.4) of g and h respectively, it is clear that it suffices to prove that, for a vertex $x \in V$ and an edge $yz \in E$ with y closer to x than z is, we have

$$f_{\chi(yz)}(x,y) \ge f_{\chi(yz)}(x,z).$$

Indeed, this holds because a geodesic from x to y ending in colour yz may be extended along the edge yz without incurring an additional colour change. □

Here we're really using the nature of colour changes to extend these paths

Claim 2.5. *Fix an integer $k \in [n]$. We then have $\mathbb{E}_k[f(u,v)] \le \mathbb{E}_k[h(u,v)] + \mathbb{E}_k[S^u(v)]$.*

Proof. We may unwrap the definitions (2.1), (2.2), and (2.4) to compute that

$$\begin{aligned}\mathbb{E}_k[f(u,v)] - \mathbb{E}_k[h(u,v)] &= \sum_{c \in \{\text{Red}, \text{Blue}\}} \left(\frac{|J_c^u(v)|}{n-k} - \frac{|I_c^u(v)|}{k} \right) \mathbb{E}_k\left[f_c(u,v)\right] \\ &\le \mathbb{E}_k\Big[S^u(v) \cdot \big| f_{\text{Red}}(u,v) - f_{\text{Blue}}(u,v) \big| \Big].\end{aligned}$$

Recalling that Observation 2.1 tells us that $|f_{\text{Red}}(x,y) - f_{\text{Blue}}(x,y)| \le 1$ for all $x, y \in V$, the claimed inequality follows. □

The definition of $S^u(v)$ can be thought of as "the thing that makes Claim 2.5 work", but there's more to it which we use in the final proof of the section.

We remark here that inequality in Claim 2.5 holds even if we condition on the values of u and v, and the proof runs exactly as above. However, the weaker statement both aligns with our notation $\mathbb{E}_k$ and suffices for our needs.

We may combine Claims 2.3, 2.4, and 2.5 to arrive at the following important corollary.

When coming up with the proof, I started with having the above three claims all rolled into one proof, and only very late in the process of writing-up did it become clear that it would be smoother to break the process up into three small steps.

Corollary 2.6. *The following holds for all $k \in [n-1]$.*

$$\mathbb{E}_k\big[f(u,v)\big] \leq \mathbb{E}_{k-1}\big[f(u,v)\big] + \mathbb{E}_k\big[S^u(v)\big].$$

To complete the proof of Lemma 2.2, it is clear from Corollary 2.6 that it suffices to show the following.

Claim 2.7. *Fix an integer $k \in [n]$. We have*

$$\mathbb{E}_k\big[S^u(v)\big] \leq \frac{1}{2}\sqrt{\frac{k(n-k)}{n}}\Big(\frac{1}{n-k}+\frac{1}{k}\Big)(1+o(1)).$$

Proof. This claim in fact holds even if we condition on v. Indeed, fix a vertex x and assume that x is in r red edges. Let u be conditioned to be at distance k from x, and let X be the random variable of how many of the red edges incident to x point towards u (that is, their other end is closer to u than x is). We can recall the definition (2.1) of the set $S^u(x)$ and compute

$$S^u(x) = \left|\frac{|J^u_{\text{Red}}(x)|}{n-k} - \frac{|I^u_{\text{Red}}(x)|}{k}\right| = \left|\frac{r-X}{n-k} - \frac{X}{k}\right|.$$

We may note that the probability that X is equal to some number i is simply the probability that independently sampling k of the edges incident to x without replacement results in i red edges, and so

$$\mathbb{P}(X=i) = \frac{\binom{r}{i}\binom{n-r}{k-i}}{\binom{n}{k}}.$$

In particular, X has a hypergeometric distribution, for which the expectation and variance are known to be

$$\mathbb{E}[X] = \frac{kr}{n} \quad \text{and} \quad \operatorname{Var}(X) = \frac{k(n-r)(n-k)r}{n(n-1)(n-2)}. \tag{2.6}$$

Indeed, write $X = Y + kr/n$, so that $\mathbb{E}[Y]=0$. We can easily compute then that

$$S^u(x) = |Y|\Big(\frac{1}{n-k}+\frac{1}{k}\Big).$$

But we may then apply Jensen's inequality to find that

$$\mathbb{E}\big[S^u(x)\big] = \Big(\frac{1}{n-k}+\frac{1}{k}\Big)\mathbb{E}\big[\,|Y|\,\big] \leq \Big(\frac{1}{n-k}+\frac{1}{k}\Big)\sqrt{\operatorname{Var}(Y)}.$$

Note that this is the only time that we condition on v and let u vary randomly, but it's important that we do this – it's exactly this power which lets us deal with the random variable $S^u(x)$.

Note that we've said absolutely nothing about the value of r. We'll just take a worst-case in a few lines' time.

You can also not apply Jensen's and just use a formula for the absolute value of Y. There is a nice such formula, and you (apparently) win an extra factor of $\sqrt{2/\pi}$ in the constant this way, but this way perhaps feels slightly less magic.

But $\mathrm{Var}(Y) = \mathrm{Var}(X)$, and so by (2.6), we find that

$$\begin{aligned}\mathbb{E}[S^u(x)] &\le \left(\frac{1}{n-k}+\frac{1}{k}\right)\sqrt{\frac{k(n-r)(n-k)r}{n(n-1)(n-2)}}\\ &= \sqrt{\frac{k(n-k)}{n}}\left(\frac{1}{n-k}+\frac{1}{k}\right)\sqrt{\frac{(n-r)r}{(n-1)(n-2)}}\\ &\le \frac{1}{2}\sqrt{\frac{k(n-k)}{n}}\left(\frac{1}{n-k}+\frac{1}{k}\right)(1+o(1)),\end{aligned}$$

where the $o(1)$ term depends only on n, as required. □

The first equality is just rearrangement. After that, note that the rightmost square root term is maximised when $r \approx n/2$, so sub that in.

With Claim 2.7 in hand, Lemma 2.2 follows immediately by induction on k. □

Replacing $1/((n-1)(n-2))$ with $1/n^2$ is the only place we pick up this $1+o(1)$ term from, so it's very easy to control.

3 Concluding remarks and open problems

While our result does make progress towards Conjecture 1.1, there is still a large gap between our bound of $O(\sqrt{n})$ and the conjectured constant. We note that improving the bound further will require some new ideas, as there are colourings of $E(Q_n)$ for which the expected minimum number of colour changes on a path from a vertex v to its antipode $\bar{v}$ is bounded below by $\Omega(\sqrt{n})$.

Note that the new ideas introduced recently by Wu and Yang to resolve the original version of the conjecture (paths not geodesics, and antipodal colourings) don't apply here, as the topology they use doesn't seem to be able to say anything about geodesics.

Indeed, colour the edges of Q_n in layers: pick an arbitrary vertex $x \in V(Q_n)$, and colour the edge xy red if it is an even distance from x, and blue if it is at an odd distance. Let $v \in V(Q_n)$ be selected uniformly at random, and let $L = d(x, v)$ be the random variable equal to the layer of v. Then L has a binomial distribution with parameter $p = 1/2$, and so (omitting floors and ceilings for convenience) it can be computed that

$$\mathbb{E}\big[\,|L - n/2|\,\big] = 2^{-n}\sum_{j=0}^{n/2}\binom{n}{j}(n-2j) = 2^{-n}n\binom{n-1}{n/2} = \sqrt{\frac{n}{2\pi}}(1+o(1)).$$

and so on average a path needs roughly $\sqrt{n}$ colour changes to get to the antipodal vertex. In combination with Theorem 1.3, this shows that the answer to Question 1.4 is $\Theta(\sqrt{n})$. It seems reasonable to suspect, but potentially difficult to prove, that colouring by layers maximises the quantity in (1.1).

Of course, when colouring by layers, any vertex from the middle layer(s) has a monochromatic path to its antipodal vertex, which is also in the middle layer(s). However, only a small proportion of vertices of Q_n have this property, and detecting this in general would seem to require ideas beyond those presented here.

Beyond just proving a stronger bound, an advantage of the techniques presented here is the possibility of generalising to more colours. In the interest of keeping the proofs here as concise as possible, we have not pursued this direction, though it seems that they would generalise to this setting. We thus suggest a possible generalisation of Conjecture 1.1 to the following.

Question 3.1. *If $E(Q_n)$ is r-coloured for some fixed $r \le n$, must there be a pair of antipodal vertices of Q_n and a geodesic path between them which changes colour at most $r-1$ times?*

I'm not sure whether I believe the claim of Question 3.1 should be true or not. (I'm also not sure whether I believe it's true for $r = 2$!)

By colouring edges of Q_n according to their direction, it is clear that Question 3.1, if true, would be the optimal result.

Finally, we can also consider other structures in Q_n. Indeed, it is natural to consider not just a path between two antipodal vertices, but one connecting all vertices: a Hamilton path.

This also seems quite difficult, as constructing a Hamilton path (never mind the colours) is much harder than constructing a geodesic, which is trivial.

Question 3.2. *Given n, what is the minimal number $h(n)$ such that, however $E(Q_n)$ is 2-coloured, there is a Hamilton path of Q_n which changes colour at most $h(n)$ times?*

We outline an argument to show that there is a colouring of $E(Q_n)$ in which every Hamilton path has at least $\Theta(2^n/n)$ colour changes. In fact, we show that any spanning tree of Q_n has at least $\Theta(2^n/n)$ monochromatic components in this colouring.

If $n = 2^k - 1$, take $C \subseteq Q_n$ to be a Hamming code (every vertex of Q_n is either in C or adjacent to exactly one element of C, and all elements of C are pairwise at distance at least 3), and colour edges red if they are incident to C and blue otherwise, then all vertices of C are in different monochromatic components. As $|C| = 2^n/(n+1)$, we thus know that any spanning tree of Q_n has at least $\Omega(2^n/n)$ monochromatic components. Taking a suitable sub-cube of Q_n allows this result to be extended to values of n not of the form $2^k - 1$.

For general spanning trees, $\Theta(2^n/n)$ monochromatic components is the truth: a result of [3] shows that there is a spanning tree T of Q_n with at least $(1 - C/n)2^n$ leaves for some constant C. Every leaf of T must share a monochromatic component with its unique neighbour vertex, and every non-leaf vertex of T is in at most two monochromatic components, and so T has at most $O(2^n/n)$ monochromatic components.

We close by remarking that it does not seem at all obvious whether one should expect the $\Omega(2^n/n)$ lower bound for $h(n)$ as in Question 3.2 to be the truth, or indeed if the value of $h(n)$ should even be $o(2^n)$.

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